

RITIK CHAUHAN

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Education

Gautam Buddha University, Greater Noida

Integrated B.Tech-M.Tech in Computer Science and Engineering

DHT Saraswati Vidya Mandir, Ghaziabad

12th Standard

Expected Graduation: June 2026

CGPA: 8.26

2021

85%

Technical Skills

- **Languages:** Python, C, C++, SQL
- **Web Frameworks & APIs:** FastAPI, REST API, Jinja2, Streamlit, Playwright
- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, Keras, CNNs, Data Augmentation, PyCaret, LangChain, GPT4All
- **NLP & Text Analytics:** TextBlob, NLTK, Pandas, Regex, OpenPyXL
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Pandas Profiling, Advanced Excel
- **Database & ORM:** SQL, SQLite, SQLAlchemy
- **Tools & DevOps:** Uvicorn, Pydantic, Git, Shell Scripting
- **Cloud Platforms:** Google Cloud Platform (GCP), Microsoft Azure

Experience

Red Wolf Info Tech Pvt Ltd, Noida

AI/ML Intern

June 2024 – Jan 2025

- Built a voice-based AI assistant for customer service using Python.
- Designed an audio preprocessing pipeline; reduced noise more than 6kHz using STFT & filters.
- Improved ASR accuracy by 18% in noisy conditions after spectral gating.
- Integrated real-time NLU backend with FastAPI, reducing response latency to (250ms)
- Developed an intent classifier (TF-IDF + Logistic Regression) with 92% test accuracy.
- Used Pydantic for validation, Uvicorn for deployment, ensuring async performance.
- Built and deployed a fully open-source, low-latency conversational pipeline.
- Maintained codebase via Git; collaborated in a 3-member agile dev team.

Omnipresent Robotics Pvt. Ltd.

Drone Tech Intern

July 2022 – August 2022

- Designed a battery health monitoring system using INA219 and ESP32 to track voltage/current and predict battery degradation.
- Integrated obstacle detection sensors (ultrasonic/LiDAR/IR) for real-time collision avoidance.
- Built a servo-based payload delivery mechanism for emergency medicine or supplies drops.
- PID loops tuned and tested on Pixhawk to enhance drone stability and flight performance.

Projects

Automated Machine Learning Web App

Python, Streamlit, Pandas, Pandas profiling, PyCaret

- Developed a Streamlit web app for automated machine learning workflows that include data upload, EDA, model training, and export.
- Integrated PyCaret for model comparison and Pandas Profiling for generating detailed EDA reports.

Web-Based Text Analysis and Sentiment Engine

Python, Playwright, NLTK, TextBlob, OpenPyXL

- Automated extraction and sentiment analysis of 100+ articles using Playwright, NLTK, and TextBlob, generating structured reports with more precision **92%** in polarity detection.
- Streamlined the entire NLP pipeline with **Pandas**, reducing manual processing time by **80%**, and output results to Excel with key metrics such as **fog index**, **complex word ratio**, and **subjectivity score**.

Crypto Trading Bot

FastAPI, Python, Binance API, SQLAlchemy, SQLite, Jinja2, Uvicorn, REST API

- Engineered a real-time crypto trading bot with **FastAPI** and **Binance API**, delivering live market data, trade history, and wallet insights via secure REST endpoints.
- Built a responsive web interface using **Jinja2**, backed by a robust **SQLAlchemy–SQLite** stack for managing user data and trade logs.
- Integrated **authentication**, secure API key handling, and automated deployment with **Uvicorn** and custom shell scripts.